

ÖMER FARUK KOÇ

AI Platform & MLOps Engineer

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SUMMARY

AI Platform and MLOps Engineer with 3+ years of production experience building and operating ML, data, and Generative AI systems in financial services. Hands-on ownership across **RAG pipelines, agentic workflows, model lifecycle, on-premises GPU inference, Kubernetes, evaluation, and observability**, spanning backend, platform, and infrastructure layers. Experienced in moving AI workloads from prototype to controlled production environments, with recent engineering work focused on **retrieval quality, LLM reliability, safe agent execution, and reproducible AI infrastructure**.

EXPERIENCE

MLOps & Analytics Engineer

Fibabanka — Analytics Center of Excellence

Mar 2023 – Mar 2026

Istanbul, Türkiye

- Re-architected sequential Oracle feature-engineering workflows into parallel, modular **dbt + Airflow** pipelines, reducing end-to-end runtime by **75%** while improving lineage, dependency visibility, and operational maintainability.
- Designed reusable **customer-level feature, training, inference, and analytics pipelines** across multiple enterprise source systems, supporting **large-scale daily production ML processing** and replacing duplicated project-specific workflows.
- Owned deployment and lifecycle management for **10 production ML models** on **MLRun (Iguazio)**, covering validation, versioning, production promotion, retraining, monitoring, and containerized FastAPI serving across Docker and Kubernetes.
- Built a **centralized feature platform** and shared dimensional models across ML use cases, replacing duplicated model-specific pipelines with reusable production data contracts and feature definitions.
- Prototyped ML and LLM workloads in Azure and migrated validated workloads to **on-premises GPU infrastructure**, serving quantized open-source models and operating local inference workloads while meeting internal data-residency and deployment constraints.
- Engineered a **call-center intelligence pipeline** spanning WhisperX transcription, timestamp alignment, speaker diarization, LLM-based analysis, structured KPI extraction, data-quality validation, and operational monitoring.
- Delivered production-oriented **RAG and document-intelligence workflows**, integrating retrieval, vector embeddings, LLM processing, structured outputs, API boundaries, and validation into internal AI applications.

TECHNICAL SKILLS

AI Platform & MLOps: MLRun, MLflow, Kubernetes, Docker, Helm, Argo CD, GitHub Actions, Azure, On-Prem AI, Model Serving, LLM Inference, Model Lifecycle, Evaluation, Monitoring

LLM & Agentic Systems: RAG, Vector Embeddings, Hybrid Retrieval, Reranking, Prompt Engineering, LangGraph, Qdrant, Human-in-the-Loop, Guardrails, MCP, Hugging Face Transformers, PyTorch, PEFT, LoRA/QLoRA, llama.cpp

Data & Distributed Systems: Apache Airflow, dbt, Apache Kafka, Redis, PostgreSQL, Oracle, Trino/Starburst, Batch and Streaming Pipelines

Backend & Observability: Python, SQL, FastAPI, Pydantic, SQLAlchemy, OpenTelemetry, Jaeger, Prometheus, Grafana

SELECTED PROJECTS

Knowledge Base RAG [GitHub]

Built a multilingual RAG platform with **vector embeddings, tenant-aware hybrid retrieval, RRF fusion, cross-encoder reranking, evidence provenance, validation controls, and citation contracts**. On a frozen **220-question reranker benchmark**, the multilingual BGE reranker reached **0.956 Cross MRR** against **0.367** for the English baseline, with **100% Recall@5** on both cross-language slices. A separate end-to-end evaluation measured **95.9% candidate evidence recall** and **0 citation-contract failures**.

Agentic Customer Service Platform [GitHub]

Built a production-oriented Agentic AI control plane with **LangGraph + PostgreSQL checkpoints** where the LLM emits an untrusted semantic proposal while deterministic services own authentication, target resolution, policy, confirmation, revalidation, and idempotent execution. Across **540 safety attempts, 0 unsafe actions reached execution**; the operational release gate passed **18/18 scenarios** and **6/6 fault classes**. A separate **100-sample real-LLM adversarial audit** recorded **0 unauthorized mutations, confirmation bypasses, or duplicate effects**.

Decision-SQL [GitHub]

Built a governed Text-to-SQL system where the model proposes a typed decision and read-only SQL while deterministic services own **SQL parsing, authorization, policy enforcement, grain safety, PostgreSQL EXPLAIN cost checks, and restricted execution**. On a frozen **180-case benchmark** across 12 synthetic database domains, the system achieved **160/180 (88.89%) governed task success** and **108/122 (88.52%) answerable runtime task success**.

ML Platform Infrastructure [GitHub]

Built and validated a reproducible Kubernetes ML platform with **MLflow, Helm, Argo CD GitOps, autoscaling, security controls, and observability** on a local kind cluster. Load tests recorded **645,809 requests with 0% errors**; saturated throughput reached **2,935 req/s**. HPA scaled **2 to 6 replicas in 71 s**, and a fresh rebuild passed **11/11 acceptance checks in ~15 minutes**.

EDUCATION

B.Sc. in Electronics Engineering, Gebze Technical University